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Period _____ Date _____

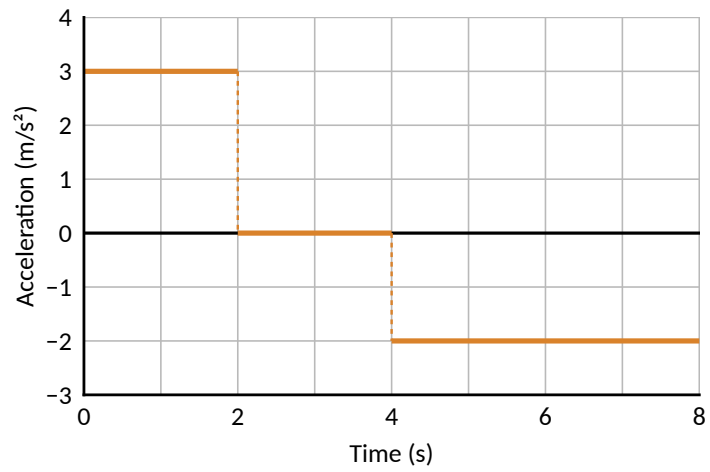
Practice Test B – Advanced Kinematics

On any problems, you must show all work, include units, round properly, and box your final answer.

Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

(25%) The acceleration-time graph below shows the motion of a car that starts from rest ($v = 0 \text{ m/s}$ at $t = 0 \text{ s}$). Sketch the corresponding velocity-time graph from $t = 0 \text{ s}$ to $t = 8 \text{ s}$.



A soccer player kicks a ball from ground level with an initial velocity of 18.0 m/s at 35.0° above the horizontal. Assume air resistance is negligible.

(25%)

a. What is the ball's hang time?

b. What is the ball's maximum height?

c. What is the ball's range?

(25%)

- How long does it take the motorcycle to reach the car's speed?
- How far behind the car is the motorcycle at that moment?
- How long does it take the motorcycle to catch up to the car, and how far has it traveled by then?

In a lab, a toy car starts from rest and is pushed by a spring-loaded plunger with a constant acceleration a over a distance d . The car then leaves the edge of a table of height h moving horizontally and lands on the floor a horizontal distance D from the edge of the table. Air resistance and friction are negligible.

(25%) Derive an expression for the plunger's acceleration a in terms of d , h , D , and physical constants, as appropriate.

Answer Key – Practice Test B

Fully worked. Algebra first, numbers last, units inside every parenthesis.

1. Sketch the v-t graph (25%)

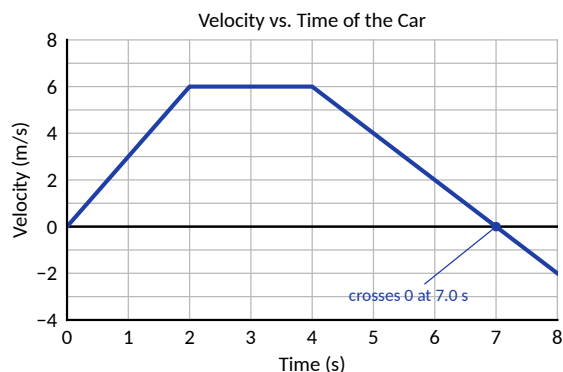
Change in velocity = area under the a-t graph. Start at $v = 0$ and add each area:

0 to 2 s: $\Delta v = (3.0 \text{ m/s}^2)(2.0 \text{ s}) = +6.0 \text{ m/s} \rightarrow v$ goes 0 to 6.0 m/s (straight line up)

2 to 4 s: $a = 0$, so v stays 6.0 m/s (flat)

4 to 8 s: $\Delta v = (-2.0 \text{ m/s}^2)(4.0 \text{ s}) = -8.0 \text{ m/s} \rightarrow v$ goes 6.0 to -2.0 m/s (straight line down)

Crosses $v = 0$ when $\frac{(6.0 \text{ m/s})}{(2.0 \text{ m/s}^2)} = 3.0 \text{ s}$ after $t = 4 \text{ s}$, so at $t = 7.0 \text{ s}$



Watch out: a constant acceleration makes a straight slanted v-t line, not a flat one. Flat v-t only where $a = 0$.

2. Soccer ball (25%)

horizontal	vertical
$a_x = 0$	$a_y = -g$
$v_{x0} = v_0 \cos \theta_0 = 14.7 \text{ m/s}$	$v_{y0} = v_0 \sin \theta_0 = 10.3 \text{ m/s}$
$\Delta x = ?$	$\Delta y = 0$ when it lands

(a) $0 = (v_0 \sin \theta_0)t - \frac{1}{2}gt^2 = t(v_0 \sin \theta_0 - \frac{1}{2}gt) \rightarrow t = \frac{2v_0 \sin \theta_0}{g}$

$t = \frac{2(18.0 \text{ m/s})(\sin 35.0^\circ)}{(9.8 \text{ m/s}^2)} = \boxed{2.11 \text{ s}}$

(b) At the top $v_y = 0$: $v_y^2 = v_{y0}^2 + 2a_y \Delta y \rightarrow 0 = v_0^2 \sin^2 \theta_0 - 2gH$

$H = \frac{v_0^2 \sin^2 \theta_0}{2g} = \frac{(18.0 \text{ m/s})^2 (\sin 35.0^\circ)^2}{2(9.8 \text{ m/s}^2)} = \boxed{5.44 \text{ m}}$

(c) $\Delta x = (v_0 \cos \theta_0)t$, substitute $t = \frac{2v_0 \sin \theta_0}{g}$:

$\Delta x = \frac{v_0^2 \sin 2\theta_0}{g} = \frac{(18.0 \text{ m/s})^2 \sin(70.0^\circ)}{(9.8 \text{ m/s}^2)} = \boxed{31.1 \text{ m}}$

3. Car and motorcycle (25%)

Car: constant $v_c = 24.0 \text{ m/s}$. Motorcycle: $v_0 = 0$, $a = 6.00 \text{ m/s}^2$. Both start at the same spot at $t = 0$.

(a) $v = v_0 + at \rightarrow v_c = 0 + at \rightarrow t = \frac{v_c}{a} = \frac{(24.0 \text{ m/s})}{(6.00 \text{ m/s}^2)} = \boxed{4.00 \text{ s}}$

(b) Car: $\Delta x_c = v_c t = (24.0 \text{ m/s})(4.00 \text{ s}) = 96.0 \text{ m}$

Motorcycle: $\Delta x_m = v_0 t + \frac{1}{2}at^2 = \frac{1}{2}(6.00 \text{ m/s}^2)(4.00 \text{ s})^2 = 48.0 \text{ m}$

gap = $96.0 \text{ m} - 48.0 \text{ m} = \boxed{48.0 \text{ m}}$

(c) Caught when the displacements are equal: $v_c t = \frac{1}{2}at^2$

$0 = t(v_c - \frac{1}{2}at) \rightarrow t = 0 \text{ (start) or } t = \frac{2v_c}{a} = \frac{2(24.0 \text{ m/s})}{(6.00 \text{ m/s}^2)} = \boxed{8.00 \text{ s}}$

$\Delta x = v_c t = (24.0 \text{ m/s})(8.00 \text{ s}) = \boxed{192 \text{ m}}$

Watch out: equal speeds (4.00 s) is not the catch-up moment. At equal speeds the gap is at its biggest.

4. Plunger launch derivation (25%), variables only

Two phases. On the table: accelerating from rest. Off the table: horizontal launch (projectile).

Fall time (vertical, $v_{y0} = 0$, $\Delta y = -h$): $-h = 0 \cdot t - \frac{1}{2}gt^2 \rightarrow t = \sqrt{\frac{2h}{g}}$

Launch speed (horizontal, $a_x = 0$): $D = v_0 t \rightarrow v_0 = \frac{D}{t} = \frac{D}{\sqrt{\frac{2h}{g}}} = D\sqrt{\frac{g}{2h}}$

Table phase (starts at rest, ends at v_0 , distance d , no time): $v_0^2 = 0^2 + 2ad \rightarrow a = \frac{v_0^2}{2d}$

Substitute v_0 : $a = \frac{D^2 \frac{g}{2h}}{2d} \rightarrow \boxed{a = \frac{gD^2}{4hd}}$

Units check: $(\text{m/s}^2)(\text{m}^2) \div (\text{m} \cdot \text{m}) = \text{m/s}^2$. The launch speed is the final speed of the table phase and the initial speed of the projectile phase; the time in the air does not apply to the table.